

-: Vocabulary For Statistics :-

Data :- Any observation that has been collected.

Statistics :- To collect data, analyze it, summarize and interpret it to form conclusion.

OR

Combination of science of uncertainty and technology and extracting information from data is called "Statistics".

Population :- Complete set of elements being studied

Sample :- Some subset of a "population".

Theme $\rightarrow$ Population $\rightarrow$ Sample

e.g: University $\rightarrow$ 4th Semester Students $\rightarrow$ 2.5+ CGPA Students

$\therefore$ Stats are mostly done over "samples"

Census :- Collecting information from every member of "population". Most time consuming process.

∴ Samples MUST be collected Randomly

Parametric Data :- Characteristics of "population."

Statistical Data :- Characteristics of "Sample"

Classification Of Quantitative & Qualitative Data :-

∴ Quantitative Data :-

1. Discrete : Countable / Finite
e.g: No. of students in class
2. Continuous : Measurements / Infinite
e.g: Weight, time, height.

Continuous Data has 2 subtypes.

1. Interval: No True zero
2. Ratio: Has True zero

* True zero: a measurement scale where 'zero' indicates complete absence of a variable being measured.

* Variables having True zero:
Blood Pressure, sugar, income 😊 😞

* Not having True zero:
admission time in a hospital,
Temperature (at Celsius & Fahrenheit)
∴ 0°C & $0^{\circ}\text{F} \rightarrow$ freezing time H_2O

∴ Qualitative Data :-

1. Nominal: Order can't be defined
e.g: measuring racism in racists,
religious people a/c to acts
2. Ordinal: Order can be defined
weather Ascending or Descending
e.g: Positions in Race,
positions in a Society

- Qualitative (Categorical) :-
Non-Numeric (e.g. Error Codes,
Age, Height system status)

- Quantitative (Numerical) :-
Numerical values (e.g. Response
zipcodes, time etc)

- Discrete (Countable) :-
e.g.: No. of Bugs

- Continuous (Measurable) :-
e.g.: weight, Time

NOTE! Basic mathematical operations
are meaningless in qualitative &
meaningless in quantitative

Methods For Simple Random Sampling :-

Random: No predictions about outcome or no specific choices in collecting any sample

① Convenient Sampling :- (not Formal)

Easy to get, sampling is done where we found ease.

e.g: in 10 rows, someone picked sample from 1st row only

② Systematic Sampling :-

Using a systematic way to collect sample.

e.g: assign numbers (1-50) to students, and say, 3's multiples from this row, evens from this row, 5's multiples from this row etc.

③ Stratified (most formal) :-

Divide the population in sub-groups regarding a particular characteristic and then collecting sample from each group.

④ Cluster Grouping Sampling :-
Regardless of any characteristic,
divide the population into sub-
groups and collect sample from
each group.

Types Of Statistics :-

Descriptive	Inferential
* Applied on both population & sample * Organised via Graphs * Uses Central Tendency (mean, median, mode) * Uses "measure of variability (variance), (standard deviation) etc.	* Applied on sample only * Using sample to make decisions about population. * Uses probability to determine how confident are we about our conclusions being correct

* Relative Frequency = $\frac{\text{Freq}}{\text{total Freq}}$

* Sum of rel. frequencies must be = 1

Q2.18:- (Qualitative Data)

"Top Broadcast Shows"

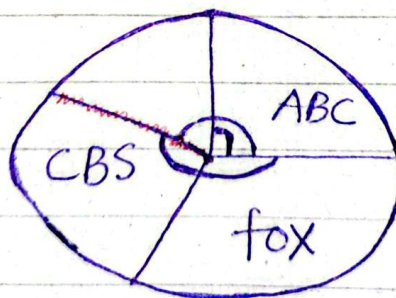
	Tally Mark	Freq	Relative Freq
ABC		5	0.25
CBS		9	0.45
fox		6	0.3
		20	1

* Pie Chart :-

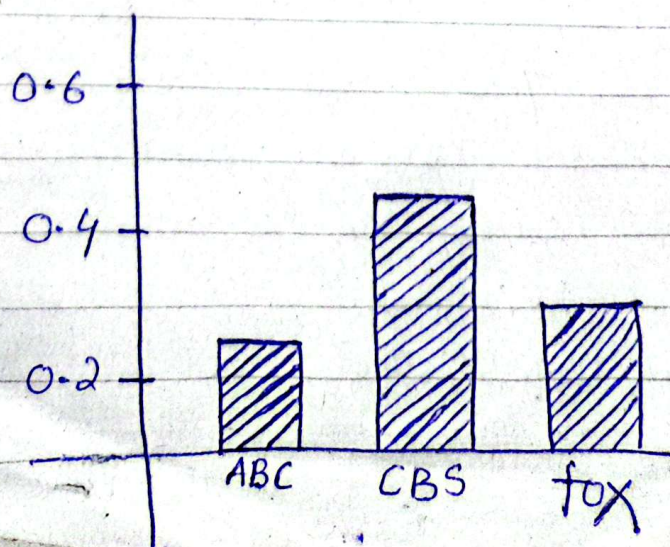
$0.25 \times 360 = 90^\circ$ (ABC)

$0.45 \times 360 = 162^\circ$ (CBS)

$0.3 \times 360 = 108^\circ$ (fox)



* Bar Graph:-



Organising Quantitative Data :-

1. Single Value Grouping :-

in a grouped data where classes represent a single value, such classes are called "single value classes" and this grouping of data $\rightarrow$ "single value grouping".

Example 2.12 :- "Trends in Televisions"

Number of Televisions	Frequency	Relative Frequency
0	1	0.02
1	16	0.32
2	14	0.28
3	12	0.24
4	3	0.06
5	2	0.04
6	2	0.04

50-houses sample was given in which lowest no. of televisions in any home was "0" and highest was "6" so these were the single value limits for our data.

2. Limit Grouping:-

Here for each class we have "lower limit" and "upper limit" also called class intervals

Example: 2.13

Table 2.6 displays number of days to maturity for 40 short-term investments, with grouping by 10's organise the data.

lowest value in data = 36

highest " " " = ~~98~~ 99

Days to Maturity	Tally Marking	Freq	Relative frequency
30-39	III	3	0.075
40-49	I	1	0.025
50-59	IIII III	8	0.2
60-69	IIII IIII	10	0.25
70-79	IIII II	7	0.175
80-89	IIII II	7	0.175
90-99	IIII	4	0.1
Total		40	1

! sometimes class-width might not be given so Apply: width =

$$\left[\frac{\text{Range}}{\text{no. of classes}} \right]$$

3. Cut-point Grouping:-

Similar to limit grouping but the first cut-point i.e. point from where we'll have to form intervals with given width, this cutpoint will be provided in the question and " - " b/w limits will be replaced by "under" term.

* Class Boundaries:-

Class	Frequency	Class Boundaries
18-25	13	17.5 - 25.5
26-33	8	25.5 - 33.5
34-41	4	33.5 - 41.5
42-49	3	41.5 - 49.5
50-57	2	49.5 - 57.5

Highly beneficial when constructing "histograms" because the space b/w bars from bar-graphs is replaced by "Correction factor" $c.f = \frac{\text{lower class limit of next class} - \text{upper class limit of previous class}}{2}$

! In "class" column, add this c.f to each upper-class limit

and subtract from each lower class limit!

* Frequency Polygon: Graph drawn

blw midpoints and frequencies, not by hands (curves) but by straight lines.

* Frequency Curve :- Hand-drawn graph blw midpoints " x " & frequencies " f ".

* Dot-Plot Diagram :-

1. Draw a horizontal axis that displays possible values of quantitative data
2. Record each observation by placing a dot over the appropriate value on horizontal axis drawn in ①.
3. Label the horizontal Axis via Name of variable.

Cumulative Frequency :-

- * less than CF: Prefix sum of normal frequency (by default)
- * greater than CF: Postfix difference from total frequency (if mentioned in exam) by normal frequency

Example: (2.13)

class intervals	class boundaries	freq	CF(<)	CF(>)
30-39	29.5-39.5	3	3	40
40-49	39.5-49.5	1	4	37
50-59	49.5-59.5	8	12	36
60-69	59.5-69.5	10	22	28
70-79	69.5-79.5	7	29	18
80-89	79.5-89.5	7	36	11
90-99	89.5-99.5	4	40	4
Totals		40		

Diagram showing arrows from the '40' in the 'Totals' row to the '40' values in the 'CF(<)' and 'CF(>)' columns, both labeled 'Equal'.

(Must be!)

* Ogive Graphs between upper class boundary and $CF(<)$ combined with lower class boundary and $CF(>)$ will give us point of intersection i.e. "Median of the data."

"Central Tendency"

1. Mean $\hat{=}$ $(\bar{x})$

* Ungroup Data : $\frac{\sum x}{n}$

* Group Data : $\frac{\sum fx}{\sum f}$

$\sum f \rightarrow$ frequency, $x \rightarrow$ midpoint

2. Median [Sort the Data First] $(\tilde{x})$

* Ungroup Data :

Odd no. of values : $\frac{(n+1)^{th}}{2}$ value

Even no. of Values : $\frac{1}{2} \left[\left(\frac{n}{2} \right)^{th} + \left(\frac{n+2}{2} \right)^{th} \right]$ value

* Group Data:

$$\tilde{x} = l + \frac{h}{f} \times \left(\frac{\sum f}{2} - CF(L) \right)$$

Where :

l = lower class boundary of median class

h = width of median class

f = frequency of median class

$CF(L)$ = $CF(L)$ of previous class to median class

Median Class = $\frac{\sum f}{2}$ th value

lying in $CF(L)$ column, that's your median class

Mode: ($\hat{x}$)

* Ungroup Data: Most common or repeated value

* Group Data:

① Mark highest frequency class as "modal class"

② Apply:

$$\hat{x} = l + h \times \left(\frac{f_m - f_1}{2f_m - f_1 - f_2} \right)$$

Where:

l = lower class boundary of modal class

h = width of modal class

f_m = frequency of modal class

f_1 = frequency of previous class to modal class

f_2 = frequency of next class to modal class

(IMP) "Measure Of Position"

* Quartiles:- [Sorted Data] Ungroup Data

- First quartile is the median part of data that lies at or below median of dataset.

- Second quartile is the "Median" of dataset.

- Third quartile, the median part of data that lies at or above median of dataset.

$$* Q_i = \frac{i(n+1)^{th} \text{ value}}{4}$$

Example: (arrange data first)

\$300 300 940 450 400
400 300 300 1050 300

Calculate Q_1, Q_2, Q_3

Solr

$$Q_1 = \frac{1(10+1)}{4}^{\text{th}} \text{ value} = \frac{11}{4}$$

$$Q_1 = 3.75^{\text{th}} = 3^{\text{rd}} \text{ value} + 0.75(4^{\text{th}} - 3^{\text{rd}})$$

$$Q_1 = 300 + 0.75(300 - 300)$$

$$Q_1 = 300$$

$$Q_2 = \frac{1}{2} \left[\left(\frac{10}{2} \right)^{\text{th}} + \left(\frac{12}{2} \right)^{\text{th}} \right] = \frac{1}{2} (5^{\text{th}} + 6^{\text{th}})$$

$$Q_2 = \frac{1}{2} (300 + 400) = 350 \text{ (median)}$$

$$Q_3 = \frac{3(11)}{4} = \frac{33}{4} = 8.25^{\text{th}} \text{ value}$$

$$Q_3 = 8^{\text{th}} + 0.25(9^{\text{th}} - 8^{\text{th}}) = 450 + 0.25(940 - 450)$$

$$Q_3 = 572.5$$

* Grouped Data:-

① Find Q_i class, $Q_i = \frac{i \leq f^{\text{th}} \text{ CF}(\angle)}{4} \text{ class}$

② Apply:

$$Q_i = l + \frac{h}{f} \left(\frac{i \leq f}{4} - \text{CF}(\angle) \right)$$

Where:

l = LCB of Q_i class

h = width of Q_i class

f = frequency of Q_i class

$\text{CF}(\angle) = \text{CF}(\angle)$ of previous class to Q_i class

Inter-Quartile Range (IQR) = $Q_3 - Q_1$

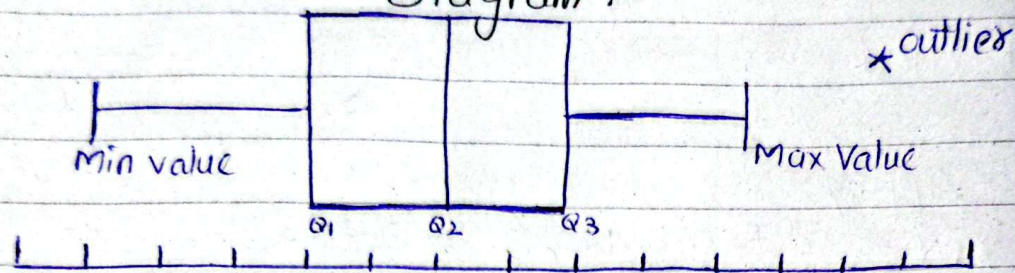
* Lower & Upper limit of dataset:

$$\text{Lower limit} = Q_1 - 1.5 \times \text{IQR}$$

$$\text{Upper limit} = Q_3 + 1.5 \times \text{IQR}$$

Values that lie below or above this range are "outliers"

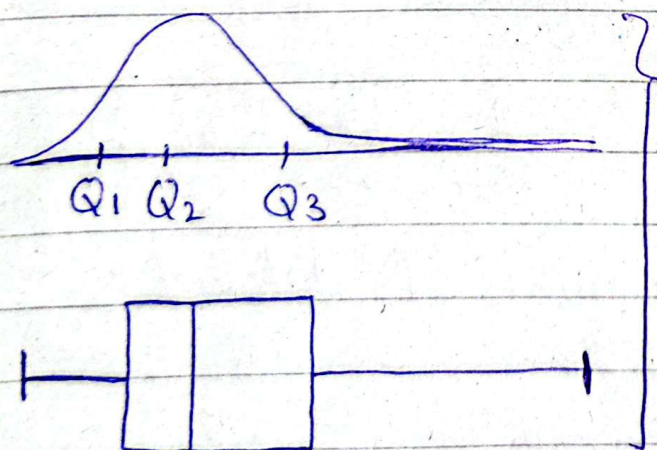
* BoxPlot / Box-and-whisker Diagram :-



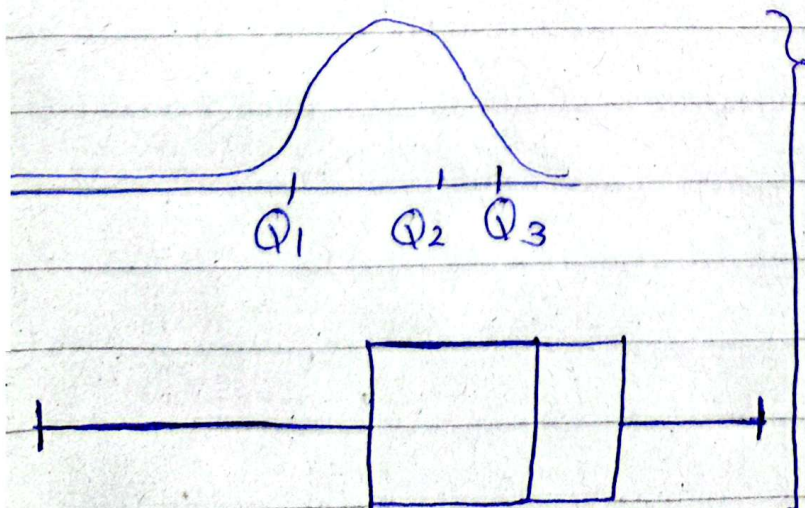
Must identify

1. Min. Value
2. Max. Value
3. Q_1 , Q_2 , Q_3
4. Outliers

Drawn above is "Uniform" in Geometrical Analysis



Right
Skewed



Left
Skewed

Measure Of Dispersion :-

* Absolute Measure of Dispersion :
used for comparisons of
variability of two or more
datasets of same units.

It includes:

- Range (Worst measure for ~~small~~ large datasets)
- Variance
- Standard Deviation

Range : Maximum Value - Minimum Value

Variance : Average squared deviation
from mean

Standard Deviation : This is the ideal
measure, $\sqrt{\text{variance}}$

Formula :-

* population variance $\Rightarrow$	$\sigma^2 = \frac{\sum (x - \bar{x})^2}{n}$	} un- group data
* Sample Variance $\Rightarrow$	$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1}$	

$$\left. \begin{array}{l} \text{* population} \\ \text{variance} \end{array} \right\} = \sigma^2 = \frac{\sum f(x - \bar{x})^2}{\sum f} \quad \left. \begin{array}{l} \text{Group} \end{array} \right\}$$

$$\left. \begin{array}{l} \text{* Sample} \\ \text{Variance} \end{array} \right\} = s^2 = \frac{\sum f(x - \bar{x})^2}{\sum f - 1} \quad \left. \begin{array}{l} \text{Data} \end{array} \right\}$$

Example: Which dataset has more variation / less consistency from given datasets

(ungroup Data)

DS1: 41, 44, 45, 47, 47, 48, 51, 53, 58, 66

DS2: 20, 37, 48, 48, 49, 50, 53, 61, 64, 70

Solr

Working on Dataset 1 ÷

x	Mean($\bar{x}$)	$(x - \bar{x})^2$
41	50	81
44		36
45		25
47		9
47		9
48		4
51		1
53		9
58		64
66		256
		<u>494</u>

$$s^2 = \frac{\sum (x - \bar{x})^2}{n-1} = \frac{494}{10-1} = 54.88$$

$$s = 7.408$$

Working on Dataset 2 :-

x	Mean ($\bar{x}$)	$(x - \bar{x})^2$
20	50	900
37		169
48		4
48		4
49		1
50		0
53		9
61		121
64		196
70		400
		<u>1804</u>

$$s^2 = \frac{\sum (x - \bar{x})^2}{n-1} = \frac{1804}{9} = 200.44$$

$$s = 14.15$$

Interpretation : Dataset 2 has more variation than dataset 1.

(Ans)

There are alternative formulas too to avoid "mean" calculation

$$\left. \begin{aligned} \sigma^2 &= \frac{\sum x^2}{n} - \left(\frac{\sum x}{n} \right)^2 \\ s^2 &= \frac{\sum x^2 - \left(\frac{(\sum x)^2}{n} \right)}{n-1} \end{aligned} \right\} \begin{array}{l} \text{Un-} \\ \text{Grouped} \\ \text{Data} \end{array}$$

$$\left. \begin{aligned} \sigma^2 &= \frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f} \right)^2 \\ s^2 &= \frac{\sum fx^2 - \left(\frac{(\sum fx)^2}{\sum f} \right)}{\sum f - 1} \end{aligned} \right\} \begin{array}{l} \text{Grouped} \\ \text{Data} \end{array}$$

Example 2.13 :-

Intervals	Boundaries	freq.	x	fx	fx^2
30 - 39	29.5 - 39.5	3	34.5	103.5	3570
40 - 49	39.5 - 49.5	1	44.5	44.5	1980
50 - 59	49.5 - 59.5	8	54.5	436	23762
60 - 69	59.5 - 69.5	10	64.5	645	41602
70 - 79	69.5 - 79.5	7	74.5	521.5	38851
80 - 89	79.5 - 89.5	7	84.5	591.5	49981
90 - 99	89.5 - 99.5	4	94.5	378	35721
Totals		40	451.5	2720	195476

$$\sigma^2 = \frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f} \right)^2 = 262.75$$

$$\sigma = 16.209$$

(Use calculator to Verify)

(Ans)

* Co-efficient of Variation :-

$$C.V = \frac{S.D}{\text{Mean}} \times 100$$

More/higher the CV, higher the variation.

Shapes Of Distribution (Geometry):-

* Symmetric: Left & Right sides are mirror images

* Skewness:

- Right-Skewed (+ve), long tail on right
- Left-Skewed (-ve), long tail on left

* kurtosis: Measures of "Peakedness"

- Leptokurtic: very peaked (thin)
- platykurtic: Very flat [image showing distributions: symmetric, +vely, -vely skewed curves]

* Stem & Leaf Diagram (Example)
12, 15, 21, 15, 30, 32, 37, 41

Stem	leaf
1	2 5 5
2	1
3	0 2 7
4	1

① Basic Probability of an Event $P(A)$:-

This is the starting point for all probability calculations it measures the likelihood of a single event A occurring:
$$P(A) = \frac{n(A)}{n(S)}$$

$\therefore n(A)$: Total no. of favorable outcomes for A

$n(S)$: Total no. of possible outcomes in a sample space
"S"

! $P(A)$ must fall b/w 0 and 1
(or 0% and 100%) !